A2 Q5

Edward Tanurkov 20401544

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5. We use the results from Q4, proving the conditions upon which we may pass a linear filter w/o distortion on an arbitrary polynomial. They will be denoted as i) and ii), same as Q4.

Let $a_0 = 1$, $a_1 = \alpha$, $a_2 = \beta$, $a_3 = \gamma$, and all other $a_j = 0$, also satisfying

i) $\sum_{j=-\infty}^{\infty} a_j = 1$ and

ii) $\sum_{j=-\infty}^{\infty} j^r a_j = 0$ for $r \geq 1$

Thus, let $m_t = \sum_{j=-\infty}^{\infty} a_j X_{t-j}$ for some time series X_t w/ a linear trend

$= (1)X_t + \alpha X_{t-1} + \beta X_{t-2} + \gamma X_{t-3}$

$= 1B^0 + \alpha B^1 + \beta B^2 + \gamma B^3$

Satisfying $\sum_{j=-\infty}^{\infty} a_j = 1$

We want to solve $\sum_{j=-\infty}^{\infty} a_j$

$= 1 + \alpha + \beta + \gamma = 1$

① $\Rightarrow 0 = \alpha + \beta + \gamma$

Satisfying $\sum_{j=-\infty}^{\infty} j^r a_j = 0$ for $r \geq 1$

We also want to solve $\sum_{j=-\infty}^{\infty} j^r a_j$ for $r = 1$ (linear trend)

$= 0^{(1)}(1) + 1^{(1)}\alpha + 2^{(1)}\beta + 3^{(1)}\gamma$

② $= \alpha + 2\beta + 3\gamma = 0$

Satisfying seasonality elimination

Given these equations, we have considered all factors but the seasonality w/ $d = 2$.

We apply the linear filter m_t assuming X_t has already been passed through by the above condition satisfactions, excepting the seasonal components at time t.

Let $m_t = X_t + \alpha X_{t-1} + \beta X_{t-2} + \gamma X_{t-3}$

$= s_t + \alpha s_{t-1} + \beta s_{t-2} + \gamma s_{t-3}$

$= (1 + \beta)s_t + (\alpha + \gamma)s_{t-1}$ $s_t = s_{t-2}$; $s_{t-1} = s_{t-3}$

In order to eliminate seasonality, we use the property that $s_t + s_{t-1} = 0$ (intuitive since seasonality must fluctuate and return to its original state at the end of its period; i.e. displacement across a period d is 0 displacement).

$\Rightarrow s_t + s_{t-1} = 0$

$\Rightarrow s_{t-1} = -s_t$

$\Rightarrow 0 = (1 + \beta)s_t - (\alpha + \gamma)s_t$

$\Rightarrow 0 = (1 + \beta) - (\alpha + \gamma)$

③ $\Rightarrow \alpha + \gamma = 1 + \beta$

We now have enough equations to solve for α, β, and γ

① $\alpha + \beta + \gamma = 0$

$\Rightarrow \alpha = -\beta - \gamma$

② $\alpha + 2\beta + 3\gamma = 0$

$\Rightarrow -\beta - \gamma + 2\beta + 3\gamma = 0$

$\Rightarrow \beta + 2\gamma = 0$

$\Rightarrow \gamma = -\frac{\beta}{2}$

③ $\alpha + \gamma = 1 + \beta$

$\Rightarrow -\beta - (-\frac{\beta}{2}) + (-\frac{\beta}{2}) = 1 + \beta$

$\Rightarrow -2\beta = 1$

$\Rightarrow \beta = -\frac{1}{2}$

② $\gamma = \frac{1}{4}$

① $\alpha = \frac{1}{2} - \frac{1}{4}$

$= \frac{1}{4}$

Sanity-checking satisfaction w/ the m_t iff. conditions.

i)

$1 + \alpha + \beta + \gamma = 1$

$1 + \frac{1}{4} - \frac{1}{2} + \frac{1}{4} = 0$ $\quad (\sum_{j=-\infty}^{\infty} a_j = 1$; i.e. i))

$1 + 0 = 1$ ✓

ii)

$\alpha + 2\beta + 3\gamma = 0$

$\frac{1}{4} - 2(\frac{1}{2}) + 3(\frac{1}{4}) = 0$ $\quad (\sum_{j=-\infty}^{\infty} j^r a_j = 0$ for $r=1$; i.e. ii))

(linear trend)

$\frac{4}{4} - \frac{2}{2} = 0$

$0 = 0$ ✓

∴ We have found a linear filter of the form $1 + \alpha B + \beta B^2 + \gamma B^3$ where $\alpha = \frac{1}{4}$, $\beta = -\frac{1}{2}$, $\gamma = \frac{1}{4}$.